

Top 50 Coding Questions - Python Solutions

1. Check Even or Odd

```
n=int(input())
print("Even" if n%2==0 else "Odd")
```

2. Check Prime Number

```
n=int(input())
prime=n>1
for i in range(2,n):
    if n%i==0:
        prime=False
        break
print("Prime" if prime else "Not Prime")
```

3. Factorial of a Number

```
n=int(input())
f=1
for i in range(1,n+1):
    f*=i
print(f)
```

4. Fibonacci Series

```
n=int(input())
a,b=0,1
for _ in range(n):
    print(a,end=" ")
    a,b=b,a+b
```

5. Reverse a Number

```
n=int(input())
rev=0
while n>0:
    rev=rev*10+n%10
    n//=10
print(rev)
```

6. Check Palindrome Number

```
n=int(input())
print("Palindrome" if str(n)==str(n)[::-1] else "Not Palindrome")
```

7. Armstrong Number

```
n=input()
sumv=sum(int(d)**3 for d in n)
print("Armstrong" if sumv==int(n) else "Not Armstrong")
```

8. Sum of Digits

```
n=int(input())
s=0
while n:
    s+=n%10
```

```
    n//=10
    print(s)
```

9. Largest of Three Numbers

```
a,b,c=map(int,input().split())
print(max(a,b,c))
```

10. GCD of Two Numbers

```
import math
a,b=map(int,input().split())
print(math.gcd(a,b))
```

11. LCM of Two Numbers

```
import math
a,b=map(int,input().split())
print(a*b//math.gcd(a,b))
```

12. Check Leap Year

```
y=int(input())
print("Leap Year" if (y%400==0 or (y%4==0 and y%100!=0)) else "Not Leap Year")
```

13. Count Vowels and Consonants

```
s=input().lower()
v=sum(1 for c in s if c in "aeiou")
c=sum(1 for ch in s if ch.isalpha() and ch not in "aeiou")
print(v,c)
```

14. Reverse a String

```
s=input()
print(s[::-1])
```

15. Check Anagram

```
a=input()
b=input()
print("Anagram" if sorted(a)==sorted(b) else "Not Anagram")
```

16. Remove Duplicates from String

```
s=input()
r=""
for ch in s:
    if ch not in r:
        r+=ch
print(r)
```

17. Find Second Largest in Array

```
arr=list(map(int,input().split()))
arr=sorted(set(arr))
print(arr[-2])
```

18. Linear Search

```
arr=list(map(int,input().split()))
key=int(input())
print("Found" if key in arr else "Not Found")
```

19. Binary Search

```
arr=list(map(int,input().split()))
key=int(input())
low,high=0, len(arr)-1
found=False
while low<=high:
    mid=(low+high)//2
    if arr[mid]==key:
        found=True
        break
    elif arr[mid]<key:
        low=mid+1
    else:
        high=mid-1
print("Found" if found else "Not Found")
```

20. Bubble Sort

```
arr=list(map(int,input().split()))
for i in range(len(arr)):
    for j in range(len(arr)-i-1):
        if arr[j]>arr[j+1]:
            arr[j],arr[j+1]=arr[j+1],arr[j]
print(arr)
```

21. Selection Sort

```
arr=list(map(int,input().split()))
for i in range(len(arr)):
    m=i
    for j in range(i+1,len(arr)):
        if arr[j]<arr[m]:
            m=j
    arr[i],arr[m]=arr[m],arr[i]
print(arr)
```

22. Insertion Sort

```
arr=list(map(int,input().split()))
for i in range(1,len(arr)):
    key=arr[i]
    j=i-1
    while j>=0 and arr[j]>key:
        arr[j+1]=arr[j]
        j-=1
    arr[j+1]=key
print(arr)
```

23. Matrix Addition

```
a=[[1,2],[3,4]]
b=[[5,6],[7,8]]
res=[[a[i][j]+b[i][j] for j in range(2)] for i in range(2)]
print(res)
```

24. Transpose of Matrix

```
a=[[1,2],[3,4]]
```

```
for row in zip(*a):
    print(row)
```

25. Count Frequency of Element

```
arr=list(map(int,input().split()))
key=int(input())
print(arr.count(key))
```

26. Check if Array is Sorted

```
arr=list(map(int,input().split()))
print("Sorted" if arr==sorted(arr) else "Not Sorted")
```

27. Merge Two Arrays

```
a=list(map(int,input().split()))
b=list(map(int,input().split()))
print(a+b)
```

28. Find Missing Number

```
arr=list(map(int,input().split()))
n=len(arr)+1
print(n*(n+1)//2-sum(arr))
```

29. Count Words in String

```
s=input().split()
print(len(s))
```

30. Remove All Spaces from String

```
s=input()
print(s.replace(" ",""))
```

31. Find Duplicate Elements in Array

```
arr=list(map(int,input().split()))
seen=set()
for x in arr:
    if x in seen:
        print(x,end=" ")
    seen.add(x)
```

32. Move All Zeros to End

```
arr=list(map(int,input().split()))
res=[x for x in arr if x!=0]+[0]*arr.count(0)
print(res)
```

33. Rotate Array Right by 1

```
arr=list(map(int,input().split()))
arr=[arr[-1]]+arr[:-1]
print(arr)
```

34. Check Palindrome String

```
s=input()
print("Palindrome" if s==s[::-1] else "Not Palindrome")
```

35. Count Number of Digits

```
n=input()
print(len(n))
```

36. Sum of Elements in Array

```
arr=list(map(int,input().split()))
print(sum(arr))
```

37. Find Minimum Element in Array

```
arr=list(map(int,input().split()))
print(min(arr))
```

38. Pattern Printing

```
n=int(input())
for i in range(1,n+1):
    print(""*i)
```

39. Power of a Number

```
a,b=map(int,input().split())
print(a**b)
```

40. Decimal to Binary

```
n=int(input())
print(bin(n)[2:])
```

41. Binary to Decimal

```
b=input()
print(int(b,2))
```

42. Check Perfect Number

```
n=int(input())
s=sum(i for i in range(1,n) if n%i==0)
print("Perfect" if s==n else "Not Perfect")
```

43. Strong Number

```
import math
n=input()
print("Strong" if sum(math.factorial(int(d)) for d in n)==int(n) else "Not Strong")
```

44. Count Even and Odd Numbers in Array

```
arr=list(map(int,input().split()))
even=sum(1 for x in arr if x%2==0)
print("Even:",even,"Odd:",len(arr)-even)
```

45. Find Intersection of Two Arrays

```
a=set(map(int,input().split()))
b=set(map(int,input().split()))
print(a & b)
```

46. Check Substring

```
s=input()
sub=input()
print("Present" if sub in s else "Not Present")
```

47. Remove Specific Character from String

```
s=input()
ch=input()
print(s.replace(ch,""))
```

48. Sum of Prime Numbers upto N

```
n=int(input())
def prime(x):
    if x<2:return False
    for i in range(2,int(x**0.5)+1):
        if x%i==0:return False
    return True
print(sum(i for i in range(2,n+1) if prime(i)))
```

49. Reverse Words in a Sentence

```
s=input().split()
print(" ".join(s[::-1]))
```

50. Two Sum Problem

```
arr=list(map(int,input().split()))
t=int(input())
for i in range(len(arr)):
    for j in range(i+1,len(arr)):
        if arr[i]+arr[j]==t:
            print(i,j)
```